

FinRiskNet: Multi-Modal Default Prediction with Risk Attribution and Test-Time Adaptation

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Abstract. Accurate estimation of the Probability of Default (PD) is central to credit risk assessment, but it is complicated by multi-source heterogeneous financial data, severe class imbalance, and non-stationary economic conditions. Existing methods tend to trade interpretability for predictive accuracy, and most are trained under a stationarity assumption that degrades when the underlying data distribution drifts. We present FinRiskNet, a multi-modal neural framework for PD prediction that is interpretable and adaptive by design. FinRiskNet fuses four complementary modalities—stable user attributes, temporal financial states, transaction-graph embeddings, and textual signals from credit reports—through a gated heterogeneous encoder coupled with a Transformer-based temporal module, and is trained with a composite objective that combines a dynamic focal term, a calibration prior, and a temporal smoothness term to counter class imbalance and non-stationarity. Two components address the limitations above: a Risk Attribution Module (RAM) yields feature-, temporal-, and modality-level explanations, with the temporal attributions validated for faithfulness, and a lightweight Selective Parameter Recalibration (SPR) strategy adapts the model to concept drift while updating only 13.7% of the parameters. On a real-world credit dataset, FinRiskNet compares favorably with strong baselines including XGBoost and adapted Transformer models, reducing Log Loss by 17.1% and raising AUROC from 0.7892 to 0.8375. Ablation and attribution-faithfulness studies confirm the contribution of each component, yielding an interpretable and drift-aware solution for credit risk analytics.

Keywords: Credit Risk Assessment · Probability of Default · Multi-Modal Fusion · Test-Time Adaptation · Explainable AI

1 Introduction

Credit risk assessment is central to modern financial services, where accurately estimating the Probability of Default (PD) underpins lending decisions, capital allocation, and regulatory compliance [4]. PD prediction is intrinsically difficult for three intertwined reasons: borrower profiles are inherently multi-modal, spanning static attributes, time-varying financial states, relational transaction patterns, and unstructured textual reports [3,32]; default events are rare, inducing severe class imbalance that biases models toward the majority class [5,23];

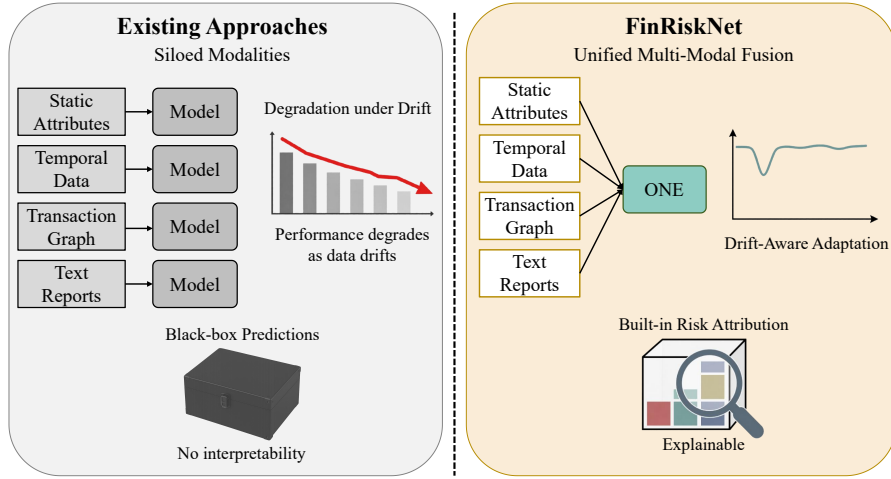


Fig. 1. Motivation for FinRiskNet. Existing methods model heterogeneous data sources in isolation, degrading under drift and offering little interpretability (left); FinRiskNet instead unifies all four modalities and provides drift-aware, interpretable predictions (right).

and macroeconomic conditions evolve continuously, causing concept drift that erodes performance over time [25]. Compounding these challenges, lending decisions are subject to regulatory scrutiny, so a model must be not only accurate but also interpretable.

Two families of methods dominate current practice. Traditional credit scoring relies on logistic regression, scorecards, and gradient-boosted tree ensembles [31,7,18] over hand-engineered tabular features; these are well understood but depend on manual feature design and treat each record independently, ignoring the temporal evolution of borrower behavior. Deep learning offers an alternative: Transformer architectures capture long-range dependencies in sequential data [34,22], graph networks model relational transaction structure [13,36], and financial language models extract risk signals from text [38,15]. Each line, however, advances a single modality in isolation.

This isolation is the central gap. Existing credit models rarely fuse tabular, temporal, relational, and textual evidence within one framework [32,6], seldom pair such fusion with optimization aimed at class imbalance and calibration, and typically handle concept drift through costly periodic retraining rather than targeted adaptation [19,29]. They also offer little built-in interpretability, despite its necessity in regulated lending [26]. Addressing these four shortcomings jointly is what credit risk modeling currently lacks.

We propose FinRiskNet, whose central idea is to treat credit risk as a single multi-modal temporal estimation problem in which heterogeneous signals are adaptively weighted rather than concatenated or modeled separately. A

learnable gating mechanism fuses the four modalities, a Transformer encoder models their temporal evolution, and a composite objective enforces calibrated, imbalance-aware predictions. An integrated Risk Attribution Module (RAM) exposes feature-, temporal-, and modality-level explanations, while a Selective Parameter Recalibration (SPR) strategy adapts the model to drift by updating only a small subset of parameters. Figure 1 contrasts this unified design with conventional isolated pipelines.

Our main contributions are summarized as follows:

- We recast credit risk modeling as a unified multi-modal temporal problem and propose FinRiskNet, in which a learnable gating mechanism adaptively fuses stable, temporal, relational, and textual signals rather than processing them in isolation, addressing the fragmentation of conventional single-modality models.
- We design a composite training objective—coupling a Dynamic Focal Loss, a macroeconomic Kullback–Leibler (KL) prior, and a temporal smoothness term—that jointly targets the class imbalance and temporal non-stationarity inherent to default prediction.
- We integrate a Risk Attribution Module that produces feature-, temporal-, and modality-level explanations, and we validate the faithfulness of its temporal attributions through perturbation analysis, addressing the transparency requirements of regulated lending.
- We introduce Selective Parameter Recalibration, a trigger-based strategy that adapts the model to concept drift while updating only 13.7% of the parameters, and we accompany it with a Lipschitz-continuity and convergence analysis that characterizes the optimization landscape.

The remainder of this paper is organized as follows. Section 2 reviews related work. Section 3 details the FinRiskNet architecture, optimization strategy, and theoretical analysis. Section 4 presents experimental results, ablation studies, training dynamics, and case study analyses. Section 5 discusses limitations, and Section 6 concludes with directions for future work.

2 Related Work

2.1 Credit Risk Modeling

Traditional credit scoring relies on logistic regression and scorecards [31], which are interpretable but limited in capturing feature interactions [7,2]. Gradient-boosted tree ensembles such as XGBoost [7] and LightGBM [18] are now the industry standard on tabular data and have been extended to specialized objectives such as uplift modeling [10]; CatBoost [27] reduces prediction shift for categorical features, and TabNet [2] adds attentive, instance-wise feature selection. These methods still require manual feature engineering and treat each record independently, ignoring how borrower states evolve over time. Deep learning alternatives for credit risk [12]—temporal convolutional and recurrent

models for sequential data [21], and variational autoencoders for latent risk representations [33]—typically operate on a single modality and lack principled handling of class imbalance or concept drift [25,5]. FinRiskNet instead addresses fusion, temporal modeling, imbalance, and drift jointly.

2.2 Multi-Modal Data Fusion in Financial Applications

Integrating heterogeneous financial data streams remains an open challenge. Architectures such as TabTransformer [16] contextualize categorical features via self-attention but lack cross-modality fusion. FT-Transformer [11] extends this line by applying Transformer layers to both numerical and categorical features, yet remains confined to a single tabular modality. Graph neural networks have been applied to transaction networks for fraud detection [24], with GraphSAGE [13] and heterogeneous graph attention networks [36] enabling scalable relational reasoning over large financial graphs. On the textual side, pre-trained language models such as FinBERT [38,15] and domain-specific BERT variants [1] have shown promise in extracting risk signals from financial documents. Multi-modal fusion strategies in related domains include early concatenation, late fusion, and cross-attention mechanisms [3]; however, prior work in credit risk has largely addressed these modalities in isolation [33,21]. In contrast, we adaptively integrate all four modalities through a gated fusion mechanism that learns complementary cross-modal interactions [32,6].

2.3 Temporal Modeling and Concept Drift

Transformer-based architectures have demonstrated superior performance over RNNs for modeling sequential data [34], with applications spanning natural language processing, time-series forecasting [40], and session-aware recommendation. In financial services, sequential modeling is particularly important for capturing evolving borrower risk profiles. Temporal Fusion Transformer (TFT) [22] combines variable selection with multi-horizon forecasting, while Autoformer [37] introduces auto-correlation mechanisms for long-range dependency modeling. However, financial data exhibits non-stationarity driven by macroeconomic cycles, regulatory changes, and market dynamics, leading to concept drift that degrades model performance over time [25]. Existing approaches to concept drift include periodic retraining, ensemble-based methods such as dynamic weighted majority [19], and online learning with adaptive regularization [29]. These strategies are often computationally expensive and may not preserve learned representations. Our Selective Parameter Recalibration (SPR) instead performs efficient, trigger-based adaptation that updates only critical components while preserving deep representations, trading off adaptation speed against stability.

2.4 Class Imbalance in Financial Prediction

Default events are inherently rare, resulting in severe class imbalance that biases conventional models toward the majority class. Classical remedies include

resampling techniques such as SMOTE [5] and its variants, cost-sensitive learning [9], and threshold-moving strategies. Focal Loss [23], originally proposed for object detection, has been widely adopted in imbalanced classification tasks by down-weighting well-classified examples. More recent approaches include class-balanced losses based on effective sample numbers [8]. In credit risk modeling, the interplay between class imbalance and non-stationarity poses a compounded challenge, as the minority class distribution itself shifts over time [25]. FinRiskNet addresses this with a Dynamic Focal Loss whose focusing parameter responds to batch-level class separability, coupled with a KL-Prior term that anchors predictions to macroeconomic base rates for better calibration.

3 Method

3.1 Risk-Centric Data Representation Learning

We present FinRiskNet, a neural framework for accurate and interpretable Probability of Default (PD) estimation. The framework integrates multi-source, heterogeneous financial data into a unified temporal risk profile for each borrower u at time t .

The data sources and representation build on four complementary feature categories, each processed by a dedicated encoder:

Stable attributes ($\mathbf{z}_u^{\text{stable}}$). Time-invariant or quasi-static features such as education level and initial occupation are encoded via embedding layers for categorical features and sliding-window standardization for continuous features:

$$\mathbf{z}_u^{\text{stable}} = \text{Emb}_{\text{cat}}(\text{CategoricalFeat}) \oplus \text{Norm}_{\text{sliding}}(\text{ContinuousFeat}) \quad (1)$$

where $\oplus$ denotes vector concatenation throughout.

Temporal state sequence ($\mathbf{X}_u^{\text{seq}}$). A time series of financial and behavioral metrics (e.g., monthly debt-service ratio, payment delinquency flags) is computed over a look-back window of length L :

$$\mathbf{X}_u^{\text{seq}} = [\mathbf{x}_{t-L}, \mathbf{x}_{t-L+1}, \dots, \mathbf{x}_t] \in \mathbb{R}^{(L+1) \times d_\tau} \quad (2)$$

where d_τ is the per-step feature dimension. All features are normalized using statistics from a trailing window to mitigate distribution shift.

Transaction graph embedding ($\mathbf{z}_u^{\text{graph}}(t)$). A dynamic user-merchant transaction graph $\mathcal{G}_t$ is periodically constructed, from which a two-layer Graph Attention Network (GAT) learns a context-aware user representation [35,13]:

$$\mathbf{z}_u^{\text{graph}}(t) = \text{GAT}_2(\mathbf{h}_u, \{\mathbf{h}_v, \forall v \in \mathcal{N}_t(u)\}; \boldsymbol{\Theta}_{\text{GAT}}) \quad (3)$$

Here, $\mathbf{h}_*$ denotes initial node features and $\mathcal{N}_t(u)$ is the set of user u 's transaction neighbors within a period ΔT .

Textual risk signal ($\mathbf{z}_u^{\text{text}}(t)$). Unstructured text from credit reports is processed by a domain-adapted pre-trained language model (LM) [38,1], with the [CLS] token embedding serving as the semantic risk summary:

$$\mathbf{z}_u^{\text{text}}(t) = \text{LM}_\phi(\text{Tokenize}(\text{Report}_u(t)))_{[\text{CLS}]} \quad (4)$$

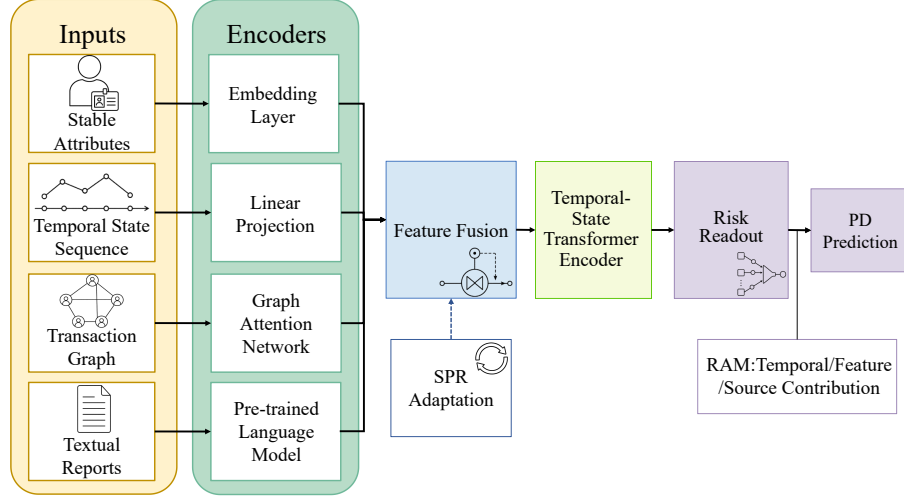


Fig. 2. Architecture of FinRiskNet. Four input modalities—stable attributes, temporal state sequences, the transaction graph, and textual reports—are processed by dedicated encoders and combined in a gated heterogeneous fusion layer. The fused sequence is contextualized by a Temporal-State Transformer Encoder and aggregated by a Hierarchical Risk Readout with attention pooling to predict PD. The Risk Attribution Module (RAM) provides temporal, feature, and source-level explanations, while Selective Parameter Recalibration (SPR) enables low-cost adaptation under concept drift.

3.2 FinRiskNet Architecture

The core architecture of FinRiskNet comprises a heterogeneous fusion encoder that models interactions among temporal dynamics, relational context, and semantic signals to produce a calibrated PD estimate. An overview of the complete architecture is presented in Figure 2.

The Heterogeneous Feature Fusion Layer aligns and adaptively combines the dynamic inputs at each time step t [3]. Let $\tilde{\mathbf{z}}_\tau(t) = \text{Linear}(\mathbf{x}_t)$ denote the projected temporal state, and let MLP denote a multi-layer perceptron. Fusion is performed via a learned gating mechanism:

$$\mathbf{g}(t) = \sigma(\mathbf{W}_g[\tilde{\mathbf{z}}_\tau(t) \oplus \mathbf{z}_u^{\text{graph}}(t) \oplus \mathbf{z}_u^{\text{text}}(t)] + \mathbf{b}_g) \quad (5)$$

$$\mathbf{z}_u^{\text{fused}}(t) = \mathbf{g}(t) \odot \text{MLP}_{\text{fuse}}([\tilde{\mathbf{z}}_\tau(t) \oplus \mathbf{z}_u^{\text{graph}}(t) \oplus \mathbf{z}_u^{\text{text}}(t)]) \quad (6)$$

The sequence of fused vectors, $\mathbf{Z}_u^{\text{fused}} = [\mathbf{z}_u^{\text{fused}}(t-L), \dots, \mathbf{z}_u^{\text{fused}}(t)]$, forms the primary input to the temporal encoder.

The Temporal-State Transformer Encoder, a multi-layer bidirectional Transformer encoder, contextualizes the fused risk state sequence to capture long-range dependencies predictive of default [34,22]:

$$\mathbf{H}_u = \text{TransformerEncoder}(\mathbf{Z}_u^{\text{fused}} + \mathbf{PE}; \boldsymbol{\Theta}_{\text{Trans}}), \quad \mathbf{H}_u \in \mathbb{R}^{(L+1) \times d_h} \quad (7)$$

where $\mathbf{PE}$ denotes the positional encoding and d_h is the hidden dimension.

The Hierarchical Risk Readout and Prediction module generates the final prediction through three components. A global sequence representation is first obtained via learned attention pooling over $\mathbf{H}_u$, then concatenated with the stable attributes and the most recent fused state:

$$\alpha_i = \frac{\exp(\mathbf{q}^\top \mathbf{H}_u[:, i])}{\sum_{j=0}^L \exp(\mathbf{q}^\top \mathbf{H}_u[:, j])}, \quad \mathbf{h}_u^{\text{global}} = \sum_{i=0}^L \alpha_i \mathbf{H}_u[:, i] \quad (8)$$

$$\mathbf{r}_u^{\text{final}} = \text{LayerNorm}(\mathbf{h}_u^{\text{global}} \oplus \mathbf{z}_u^{\text{stable}} \oplus \mathbf{z}_u^{\text{fused}}(t)) \quad (9)$$

$$\hat{y}_u = \text{PD}(t + H \mid \mathcal{I}_t) = \sigma(\text{MLP}_{\text{pred}}(\mathbf{r}_u^{\text{final}})) \quad (10)$$

where H is the prediction horizon, $\mathcal{I}_t$ is the multi-modal information set available up to time t (formally defined in Definition 1), $\mathbf{q} \in \mathbb{R}^{d_h}$ is a learnable attention query, and $\sigma(\cdot)$ is the logistic sigmoid.

For interpretability, the Integrated Risk Attribution Module (RAM) uses internal model signals to provide multi-level explanations [26,28,30]. Temporal attention scores α_i identify critical historical time steps; feature attribution computes Integrated Gradients with respect to raw input features $\mathbf{x}_t$ and text tokens; and source contribution decomposes the final logit score into additive contributions from the graph ($\mathbf{z}_u^{\text{graph}}$), text ($\mathbf{z}_u^{\text{text}}$), and temporal ($\mathbf{X}_u^{\text{seq}}$) pathways using a linear approximation of the fusion MLP.

3.3 Optimization for Imbalanced and Non-Stationary Data

The composite loss addresses class imbalance and encourages calibrated predictions:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{DFL}} + \lambda \mathcal{L}_{\text{KL-Prior}} + \beta \mathcal{L}_{\text{TempSmooth}} \quad (11)$$

where $\lambda > 0$ and $\beta > 0$ are scalar weights balancing the calibration and temporal-smoothness terms against the primary classification loss.

Dynamic Focal Loss (DFL). The first term extends Focal Loss [23] with an adaptive focusing parameter γ that adjusts based on estimated class separability within each mini-batch:

$$\mathcal{L}_{\text{DFL}} = -\frac{1}{N} \sum_{i=1}^N w_{y_i} \cdot (1 - \hat{p}_i)^{\gamma(\kappa)} \cdot [y_i \log(\hat{p}_i) + (1 - y_i) \log(1 - \hat{p}_i)] \quad (12)$$

Here, N is the mini-batch size, $\hat{p}_i \in [0, 1]$ is the predicted default probability for sample i , $y_i \in \{0, 1\}$ is its label, and $w_{y_i} > 0$ is a class-dependent weight. The focusing parameter $\gamma(\kappa)$ is a monotonically increasing function of the batch-level class separability κ , so that γ grows when the model is overconfident on negative samples, forcing increased focus on hard negatives.

KL divergence with macroeconomic prior ($\mathcal{L}_{\text{KL-Prior}}$). The second term calibrates predictions by softly aligning the output distribution with a macroeconomic prior $\text{Beta}(\alpha_{\text{econ}}, \beta_{\text{econ}})$, estimated from historical default rates:

$$\mathcal{L}_{\text{KL-Prior}} = \frac{1}{N} \sum_{i=1}^N D_{\text{KL}}(\text{Beta}(\hat{p}_i \cdot \nu + 1, (1 - \hat{p}_i) \cdot \nu + 1) \parallel \text{Beta}(\alpha_{\text{econ}}, \beta_{\text{econ}})) \quad (13)$$

The concentration parameter $\nu > 0$ controls the regularization strength, and $\alpha_{\text{econ}}, \beta_{\text{econ}} > 0$ are the shape parameters of the macroeconomic prior estimated from historical default rates [20].

Temporal smoothness regularization ($\mathcal{L}_{\text{TempSmooth}}$). The third term encourages coherent risk trajectories by penalizing excessive volatility in sequential PD predictions for a given user:

$$\mathcal{L}_{\text{TempSmooth}} = \frac{1}{N(L-1)} \sum_{i=1}^N \sum_{l=1}^{L-1} (\hat{y}_u^{(t_{l+1})} - \hat{y}_u^{(t_l)})^2 \quad (14)$$

Here, $\hat{y}_u^{(t_l)} = \sigma(\text{MLP}_{\text{pred}}(\mathbf{H}_u[:, l]))$ is the per-step default probability obtained by applying the shared prediction head to the contextualized hidden state at sequence step t_l , yielding a risk trajectory over the look-back window that this term regularizes for temporal coherence.

3.4 Dynamic Model Adaptation to Concept Drift

To maintain performance under macroeconomic shifts, FinRiskNet employs a lightweight, trigger-based recalibration strategy. We monitor the prediction distribution shift (PDS) on a recent validation holdout:

$$\text{PDS}(t) = \text{JS}(P_{\mathcal{D}_{\text{val}}}(\hat{y}_{\theta_t}) \parallel P_{\mathcal{D}_{\text{val}}}(\hat{y}_{\theta_{t-\Delta}})) \quad (15)$$

where $\text{JS}(\cdot \parallel \cdot)$ is the Jensen–Shannon divergence and $P_{\mathcal{D}_{\text{val}}}(\cdot)$ is the predictive distribution over the validation holdout. If $\text{PDS}(t) > \tau_{\text{drift}}$, a concept drift alarm triggers selective parameter recalibration (SPR) [29,19]. This involves fine-tuning only the parameters of the prediction head MLP_{pred} , the fusion gate $(\mathbf{W}_g, \mathbf{b}_g)$, and the final layer of the temporal encoder using the most recent data. This enables rapid adaptation while preserving deep feature representations.

3.5 Theoretical Analysis

We provide a formal characterization of the FinRiskNet optimization landscape. Proposition 1 and Theorem 1 establish standard regularity and convergence guarantees for the composite loss, and Remark 1 characterizes the stability of Selective Parameter Recalibration under concept drift. We give proof sketches below.

Definition 1 (Multi-Modal Credit Risk Estimation). *Given a borrower u observed over a look-back window $[t-L, t]$, let*

$$\mathcal{I}_u(t) = \{\mathbf{z}_u^{\text{stable}}, \mathbf{X}_u^{\text{seq}}, \mathbf{z}_u^{\text{graph}}(t), \mathbf{z}_u^{\text{text}}(t)\}$$

denote the complete multi-modal information set. The objective is to learn a parameterized mapping $f_\theta : \mathcal{I}_u(t) \mapsto [0, 1]$ that minimizes the expected composite risk:

$$\theta^* = \arg \min_{\theta \in \Theta} \mathbb{E}_{(u,t) \sim \mathcal{D}} [\mathcal{L}_{\text{total}}(f_\theta(\mathcal{I}_u(t)), y_u)], \quad (16)$$

where $y_u \in \{0, 1\}$ is the binary default indicator and $\mathcal{L}_{\text{total}}$ is the composite loss defined in Section 3.

Proposition 1 (Lipschitz Continuity of the Composite Loss). *Assume that the gating function $\sigma(\cdot)$ and the fusion MLP are Lipschitz continuous with constants C_σ and C_{fuse} , respectively, and that all input modalities are bounded, i.e., $\|\mathbf{z}^{(m)}\|_2 \leq B$ for each modality m . Then $\mathcal{L}_{\text{total}}(\theta)$ is Lipschitz continuous with respect to θ , with constant*

$$C_{\text{total}} \leq C_{\text{DFL}} + \lambda C_{\text{KL}} + \beta C_{\text{TS}}, \quad (17)$$

where C_{DFL} , C_{KL} , and C_{TS} are the Lipschitz constants of the Dynamic Focal Loss, KL-Prior, and Temporal Smoothness terms, respectively.

Proof (Proof Sketch). The sigmoid output $\hat{p}_i = \sigma(\text{logit}_i)$ is Lipschitz with constant $1/4$. Since the Dynamic Focal Loss is a composition of the focal weighting $(1 - \hat{p}_i)^{\gamma(\kappa)}$ and the cross-entropy, both Lipschitz on the clamped domain $[\epsilon, 1 - \epsilon]$, $\mathcal{L}_{\text{DFL}}$ inherits Lipschitz continuity. The KL divergence between Beta distributions with bounded concentration parameters is similarly Lipschitz, and the quadratic temporal smoothness term is Lipschitz with respect to the model parameters via the chain rule. As a non-negative weighted sum of Lipschitz functions, the composite loss is itself Lipschitz, with constant bounded by the weighted sum of the individual constants as stated.

Theorem 1 (Convergence of Stochastic Optimization). *Let $\{\theta_k\}_{k=0}^K$ denote the parameter sequence produced by stochastic gradient descent with learning rate $\eta_k = \eta_0/\sqrt{k+1}$ on the composite loss $\mathcal{L}_{\text{total}}$. Under the assumptions that (i) $\mathcal{L}_{\text{total}}$ is L -Lipschitz smooth, (ii) the stochastic gradient has bounded variance $\mathbb{E}[\|\nabla \mathcal{L}_{\text{total}}(\theta_k; \xi_k) - \nabla \mathcal{L}_{\text{total}}(\theta_k)\|^2] \leq s^2$, and (iii) $\mathcal{L}_{\text{total}}$ is bounded below by $\mathcal{L}^*$, then*

$$\min_{0 \leq k \leq K} \mathbb{E}[\|\nabla \mathcal{L}_{\text{total}}(\theta_k)\|^2] \leq \mathcal{O}\left(\frac{\mathcal{L}_{\text{total}}(\theta_0) - \mathcal{L}^* + s^2}{\sqrt{K+1}}\right). \quad (18)$$

Proof (Proof Sketch). Under the stated assumptions of L -smoothness, bounded gradient variance, and a lower-bounded objective, standard non-convex stochastic optimization analysis with the diminishing step size $\eta_k = \eta_0/\sqrt{k+1}$ yields the stated $\mathcal{O}(1/\sqrt{K})$ rate to a first-order stationary point. This confirms that the composite objective remains stable to optimize despite coupling three heterogeneous loss terms.

Remark 1 (Stability under Selective Parameter Recalibration). When SPR is triggered, only a subset of parameters $\theta_{\text{SPR}} \subset \theta$ (the fusion gate, the final Transformer layer, and the prediction head) is updated. Since the remaining parameters $\theta \setminus \theta_{\text{SPR}}$ are frozen, the effective optimization landscape is restricted to a lower-dimensional subspace. This restriction preserves the learned deep feature representations and limits the magnitude of parameter perturbation, thereby bounding the post-recalibration performance deviation:

$$|\mathcal{L}_{\text{total}}(\theta') - \mathcal{L}_{\text{total}}(\theta)| \leq C_{\text{total}} \cdot \|\theta'_{\text{SPR}} - \theta_{\text{SPR}}\|_2, \quad (19)$$

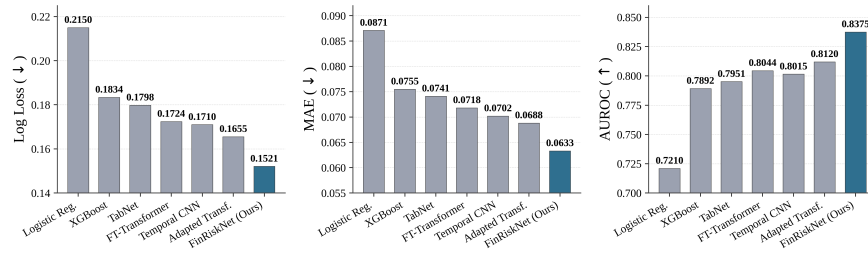


Fig. 3. Performance comparison on the PD prediction task. Arrows indicate metric direction: $\downarrow$ denotes lower is better, $\uparrow$ denotes higher is better. The highlighted (darker) bar denotes our method (FinRiskNet); lighter bars are baselines. FinRiskNet achieves the best Log Loss (0.1521), MAE (0.0633), and AUROC (0.8375) across all evaluated models, supporting the effectiveness of its multi-modal fusion design.

where θ' denotes the recalibrated parameters. This bound formalizes the empirical observation in Section 4 that SPR achieves performance comparable to full fine-tuning while modifying substantially fewer parameters.

4 Experiments

4.1 Experimental Setup

We evaluate FinRiskNet on Probability of Default (PD) prediction, using a 12-month observation window to predict default within the subsequent 12 months.

The experiments draw on the following datasets. We use a proprietary dataset from a consumer lender ($\approx 500,000$ records over 36 months, 3.2% default rate) with full multi-modal features—structured attributes, monthly behavioral series, transaction-graph snapshots, and credit-report text. We additionally evaluate on two public benchmarks, Give Me Some Credit [17] (150,000 records, 10.5% default) and Taiwan Credit [39] (30,000 records, 22.1% default). On the public sets, graph and text modalities are approximated from derived features under identical splits for fair comparison.

We use 60%/20%/20% train/validation/test splits and report Log Loss, Mean Absolute Error (MAE), and Area Under the Receiver Operating Characteristic curve (AUROC). Significance over the strongest baseline is assessed by a two-sided paired t -test ($p < 0.05$) over five seeded runs.

4.2 Main Results: Probability of Default Prediction

As shown in Table 1 and Figure 3, FinRiskNet attains the best results across all metrics. On the proprietary dataset it reduces Log Loss by 17.1% relative to the XGBoost baseline and consistently outperforms the stronger neural baselines, including FT-Transformer and the Adapted Transformer; all improvements over the best baseline are statistically significant ($p < 0.05$, paired t -test over five

Table 1. Main results for the PD prediction task on the proprietary dataset (top) and public benchmarks (bottom). Best results are in **bold**. $\downarrow$ indicates lower is better; $\uparrow$ indicates higher is better. † denotes statistically significant improvement over the best baseline ($p < 0.05$, paired t -test, 5 runs). Mean $\pm$ std reported.

Model	Log Loss $\downarrow$	MAE $\downarrow$	AUROC $\uparrow$
<i>Proprietary Dataset</i>			
Logistic Regression [31]	0.2150 $\pm$ 0.0012	0.0871 $\pm$ 0.0008	0.7210 $\pm$ 0.0015
XGBoost [7]	0.1834 $\pm$ 0.0009	0.0755 $\pm$ 0.0006	0.7892 $\pm$ 0.0011
TabNet [2]	0.1798 $\pm$ 0.0011	0.0741 $\pm$ 0.0007	0.7951 $\pm$ 0.0013
FT-Transformer [11]	0.1724 $\pm$ 0.0010	0.0718 $\pm$ 0.0007	0.8044 $\pm$ 0.0012
Temporal CNN [21]	0.1710 $\pm$ 0.0010	0.0702 $\pm$ 0.0007	0.8015 $\pm$ 0.0014
Adapted Transformer	0.1655 $\pm$ 0.0008	0.0688 $\pm$ 0.0006	0.8120 $\pm$ 0.0010
FinRiskNet (Ours)	0.1521$^\dagger \pm$ 0.0007	0.0633$^\dagger \pm$ 0.0005	0.8375$^\dagger \pm$ 0.0009
<i>Give Me Some Credit (Public)</i>			
XGBoost [7]	0.2241 $\pm$ 0.0014	0.0923 $\pm$ 0.0009	0.8512 $\pm$ 0.0013
FT-Transformer [11]	0.2108 $\pm$ 0.0012	0.0871 $\pm$ 0.0008	0.8634 $\pm$ 0.0011
FinRiskNet (Ours)	0.1973$^\dagger \pm$ 0.0010	0.0814$^\dagger \pm$ 0.0007	0.8791$^\dagger \pm$ 0.0009
<i>Taiwan Credit Dataset (Public)</i>			
XGBoost [7]	0.4312 $\pm$ 0.0021	0.1534 $\pm$ 0.0013	0.7823 $\pm$ 0.0017
FT-Transformer [11]	0.4189 $\pm$ 0.0019	0.1482 $\pm$ 0.0011	0.7941 $\pm$ 0.0015
FinRiskNet (Ours)	0.3944$^\dagger \pm$ 0.0016	0.1391$^\dagger \pm$ 0.0010	0.8187$^\dagger \pm$ 0.0013

seeds). The same ordering holds on the Give Me Some Credit and Taiwan Credit benchmarks, indicating that the gains are not specific to the proprietary data. We attribute them to the gated heterogeneous fusion of temporal, relational, and textual signals together with the imbalance- and drift-aware composite loss [14,25].

4.3 Detailed Ablation Studies

We conduct ablation studies to validate the contribution of each key component in FinRiskNet. Experiments systematically remove or alter specific modules while keeping all other settings identical to the full model. Results are reported on the hold-out test set for PD prediction.

Table 2 reports the detailed ablations. On the architecture side, replacing the gated fusion layer with plain concatenation causes the largest drop, confirming the value of the learnable gating, while substituting the Transformer with a gated recurrent unit (GRU) or removing the hierarchical readout degrades performance more modestly. Among data sources, removing the transaction-graph embeddings hurts most, followed by the textual signal and the stable attributes, indicating that relational and semantic cues add complementary predictive value. Among loss terms, the Dynamic Focal Loss contributes most—replacing it with standard cross-entropy causes the largest drop—whereas the KL

Table 2. Detailed ablation studies on the hold-out test set, grouped by architecture, data modality, and loss component. The full model is the common reference. Mean $\pm$ std over 5 runs; † denotes a statistically significant difference from the full model ($p < 0.05$, paired t -test). $\downarrow/\uparrow$: lower/higher is better.

Model Variant	Log Loss $\downarrow$	MAE $\downarrow$	AUROC $\uparrow$
FinRiskNet (Full)	0.1521$\pm$0.0007	0.0633$\pm$0.0005	0.8375$\pm$0.0009
<i>Architecture</i>			
w/o Hetero. Fusion (Direct Concat)	0.1625 $\pm$ 0.0015	0.0689 $\pm$ 0.0010	0.8224 $^\dagger \pm$ 0.0018
w/o Transformer (Replace w/ GRU)	0.1593 $\pm$ 0.0013	0.0661 $\pm$ 0.0009	0.8280 $^\dagger \pm$ 0.0015
w/o Hier. Readout (Global only)	0.1558 $\pm$ 0.0010	0.0647 $\pm$ 0.0007	0.8315 $^\dagger \pm$ 0.0012
<i>Data Modality</i>			
w/o Graph Embedding ($\mathbf{z}_u^{\text{graph}}$)	0.1579 $\pm$ 0.0012	0.0665 $\pm$ 0.0009	0.8261 $^\dagger \pm$ 0.0016
w/o Textual Signal ($\mathbf{z}_u^{\text{text}}$)	0.1555 $\pm$ 0.0011	0.0652 $\pm$ 0.0008	0.8298 $^\dagger \pm$ 0.0014
w/o Stable Attributes ($\mathbf{z}_u^{\text{stable}}$)	0.1542 $\pm$ 0.0009	0.0644 $\pm$ 0.0006	0.8320 $^\dagger \pm$ 0.0012
<i>Loss Component</i>			
w/o DFL (Use Standard BCE)	0.1560 $\pm$ 0.0011	0.0655 $\pm$ 0.0008	0.8310 $^\dagger \pm$ 0.0013
w/o KL-Prior Reg. ($\lambda = 0$)	0.1538 $\pm$ 0.0008	0.0642 $\pm$ 0.0006	0.8332 $^\dagger \pm$ 0.0011
w/o Temporal Smooth. ($\beta = 0$)	0.1530 $\pm$ 0.0008	0.0639 $\pm$ 0.0006	0.8340 $^\dagger \pm$ 0.0010

Table 3. Ablation study on dynamic adaptation to concept drift. Mean $\pm$ std over 5 runs. $\downarrow/\uparrow$: lower/higher is better. Params (%) denotes the proportion of model parameters updated during adaptation; Time is the adaptation cost relative to full fine-tuning ($1.0\times$).

Adaptation Strategy	Log Loss $\downarrow$	MAE $\downarrow$	AUROC $\uparrow$	Params (%) $\downarrow$	Time $\downarrow$
Full Model + SPR (Ours)	0.1505$\pm$0.0007	0.0620$\pm$0.0005	0.8410$\pm$0.0009	13.7	0.31 $\times$
No Adaptation (Static)	0.1689 $\pm$ 0.0016	0.0712 $\pm$ 0.0011	0.8185 $\pm$ 0.0019	0	—
Full Fine-Tuning	0.1510 $\pm$ 0.0009	0.0623 $\pm$ 0.0006	0.8405 $\pm$ 0.0011	100	1.0 $\times$
LP-FT (head only)	0.1555 $\pm$ 0.0011	0.0651 $\pm$ 0.0008	0.8328 $\pm$ 0.0014	2.4	0.15 $\times$

prior and temporal smoothness terms mainly improve calibration and trajectory coherence.

For the dynamic adaptation ablation, we simulate concept drift using data from a subsequent economic quarter. As Table 3 shows, the static model degrades sharply, whereas Selective Parameter Recalibration (SPR) nearly matches full fine-tuning (AUROC 0.8410 vs. 0.8405) while updating only 13.7% of the parameters at 0.31 $\times$ the cost. Restricting updates to the prediction head alone (LP-FT) is cheaper but leaves a clear gap, showing that the fusion gate must be adapted alongside the head. SPR thus gives the best quality–efficiency trade-off.

4.4 Training Dynamics and Loss Analysis

Training over 50 epochs is stable: the composite loss decreases smoothly, with the Dynamic Focal Loss dominating early optimization and the KL-prior and

smoothness terms acting as steady regularizers that aid calibration. When a prediction distribution shift is detected, SPR recovers validation performance within a few epochs of fine-tuning, confirming it as a practical alternative to full retraining [25,19].

4.5 Case Study and Model Attribution Analysis

To illustrate the interpretability afforded by FinRiskNet, we examine a case study with the Risk Attribution Module (RAM). For a user assigned a high predicted default probability of 0.89, the attribution decomposes the decision into three complementary signals. The temporal attention concentrated on a period of repeated late payments two months earlier, the feature attributions singled out the debt-service ratio and the number of recent credit inquiries in those months, and the textual pathway accounted for roughly 40% of the score, picking up phrases such as “overdraft fees charged” from the credit report. Transaction-graph embeddings contributed a further 30%, reflecting links to merchants in high-risk categories. Such an explanation gives risk analysts a concrete, auditable account of why a borrower is flagged, which is essential for meeting the documentation requirements that regulated lending imposes on automated decisions. This lets an analyst trace a flagged decision back to specific behavioral, relational, and textual evidence rather than accepting an opaque score.

We further assess attribution faithfulness. To verify that the RAM attributions reflect genuinely influential inputs rather than spurious signals, we conduct a perturbation-based faithfulness analysis on the temporal pathway. Guided by the RAM temporal importance scores, we remove the most influential visits and measure the resulting change in AUROC, comparing against a random-removal control. As reported in Table 4, removing the single most important visit identified by RAM degrades AUROC by 4.18%, and removing the top-two visits by 7.06%, whereas removing a randomly selected visit causes only a 0.91% change. The order-of-magnitude gap between RAM-guided and random removal confirms that the attributions localize the truly predictive signal. Retaining only the top-two visits incurs a modest 2.23% change, indicating that a compact subset of interactions captures most of the predictive evidence—a property valuable for concise, auditable risk explanations.

5 Limitations

While FinRiskNet performs well, we acknowledge several limitations. First, although we validate on two public benchmarks, the full multi-modal architecture (including transaction graph and text modalities) is evaluated primarily on proprietary data; graph and text features are approximated on public datasets due to unavailability of raw reports. Future work should evaluate on datasets with richer modality coverage. Second, the SPR adaptation mechanism relies on a manually set threshold τ_{drift} for triggering recalibration; an adaptive or

Table 4. Faithfulness of RAM temporal attributions. AUROC relative change (%) under RAM-guided versus random perturbation of visits. Larger degradation under RAM-guided removal indicates more faithful attribution.

Perturbation Strategy	AUROC Relative Change (%)
Remove Most Important Visit (top-1)	−4.18
Remove Top-2 Important Visits	−7.06
Remove Random Visit (control)	−0.91
Keep Only Top-2 Visits	−2.23

learned threshold could improve robustness. Third, the computational cost of the full multi-modal pipeline (GAT + language model + Transformer) is higher than tree-based baselines, which may limit deployment in low-latency settings; model distillation or pruning are natural avenues for improvement. Finally, the KL-Prior regularization requires estimation of macroeconomic base rates, which may not always be available in real-time production environments.

6 Conclusion

We presented FinRiskNet, a multi-modal framework that casts Probability of Default (PD) estimation as a unified, interpretable, and drift-aware learning problem. A gated heterogeneous encoder fuses static, temporal, relational, and textual signals; a composite loss targets class imbalance and temporal non-stationarity; a Risk Attribution Module exposes feature-, temporal-, and modality-level explanations; and Selective Parameter Recalibration adapts the model to concept drift at low cost. On the PD task FinRiskNet lowers Log Loss by 17.1% and raises AUROC from 0.7892 to 0.8375 over a strong XGBoost baseline, with ablations confirming each component’s role and a perturbation-based study validating the faithfulness of the temporal attributions. Future work will pursue more efficient architectures for real-time scoring, extend the fusion paradigm to further financial data sources [32,6], and explore federated learning for privacy-preserving training across institutions.

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